

Unit 3.1 - Approximation and Error

NOTE: Much of this content was taken from Heath's *Scientific Computing*. Slides available [here](#).

1 General Strategy

In mathematical and scientific computing, approximation plays a crucial role in solving complex problems. Complexity in problems can arise from various factors, including the nature of the variables, the dimensionality, and the type of equations involved. However, one fundamental observation in approximation is that the solution to a similar but simpler problem will often yield insights into the solution of the more complicated problem. Thus, common strategy is to simplify the problem by transforming it into a more manageable form. By understanding the relationships between different types of problems, we can make educated guesses about their solutions. The following simplifications illustrate how solutions can be derived from simpler analogs:

- **Continuous to Discrete:** In numerical analysis, continuous problems are often discretized for computation. The finite difference method, for example, converts differential equations into algebraic equations by approximating derivatives with differences.
- **Discrete to Continuous:** Conversely, one can also approximate discrete data with continuous models. Techniques such as interpolation and regression allow us to create continuous functions that closely fit a set of discrete points.
- **Differential to Algebraic:** In certain cases, differential equations can be transformed into algebraic equations through various techniques, such as discretization or using integral transforms. This makes them easier to solve with numerical methods.
- **Nonlinear to Linear:** Many nonlinear problems can be approximated by linear models, particularly near equilibrium points. This linearization provides a useful approach for analyzing the behavior of complex systems.

1.1 Example: Computing the Surface Area of the Earth

To compute the surface area of the Earth, we can use the formula:

$$A = 4\pi r^2 \tag{1}$$

where r is the radius of the Earth. This computation involves several important assumptions that make it a simpler version of the “true problem” of calculating the surface area of the Earth:

- **Modeling Earth as a Sphere:** For simplicity, we idealize the shape of the Earth as a perfect sphere, despite its actual oblate spheroid shape (plus mountains and valleys).
- **Empirical Measurements:** The value for the radius r is an average that is based on imperfect empirical measurements and previous computations, typically around 6,371 kilometers.
- **Truncating π :** The value of π is an irrational number that requires truncating an infinite process. For practical computations, we use a finite approximation, such as $\pi \approx 3.14159$.
- **Finite Precision Arithmetic:** The mathematical operations in the formula involve finite precision arithmetic, which can introduce small errors in the final result, especially when dealing with very large or very small numbers.

Additionally, if we used this approximate calculation of the Earth’s area in a later computation (which would also have its own sources of error), the error would propagate throughout all of the computations. Sometimes, we can be lucky and these errors can offset, but this is usually not the case.

1.2 Sources of Error

Errors in scientific and computational models can arise from various sources. These sources generally fall into three main categories, which we will discuss below.

1.2.1 Errors in the Inputs

- **Measurement precision:** Inputs to the model often come from physical measurements, which have limited precision. Errors arise due to the inherent limitations of the measuring instruments.
 - Example: A ruler may only measure length to the nearest millimeter, introducing a small but non-zero error in any input taken from measurements.
 - Example: A thermometer that measures temperature only to the nearest degree Celsius.

1.3 Errors in the Model

- **Simplifications:** To make models computationally feasible, complex systems are often simplified, which can introduce errors.
 - Example: Modeling Earth’s shape as a perfect sphere for ease of calculation, when in reality it is an oblate spheroid.
- **Discrete approximation:** Continuous mathematical problems (e.g., differential equations) are often approximated by discrete methods for numerical computation, which can introduce approximation errors.
 - Example: Approximating derivatives using finite differences introduces a small error, depending on the step size used.

1.4 Errors Due to Floating Point Precision

- **Finite precision:** Computers represent numbers using a finite number of bits, which leads to rounding errors when working with real numbers.
 - Example: The result of adding a very large number to a very small one may be inaccurate, as the smaller number may be lost due to rounding.
- **Floating point operations:** Arithmetic operations in floating point numbers are subject to rounding errors due to limited precision.
 - Example: Repeated additions or subtractions can accumulate rounding errors, leading to significant deviations from the true result.

2 Mathematics of Error

It is often necessary for us to be able to quantify how much error our simplifications and approximations introduce. Two commonly used definitions are absolute error and relative error.

2.1 Absolute Error

The absolute error measures the difference between the true value and the approximated value. It is defined mathematically as:

$$E_a = |x_{\text{true}} - x_{\text{approx}}| \quad (2)$$

where:

- E_a is the absolute error
- x_{true} is the true value
- x_{approx} is the approximated value

2.2 Relative Error

The relative error provides a measure of the size of the absolute error in relation to the true value. It is defined as:

$$E_r = \frac{|x_{\text{true}} - x_{\text{approx}}|}{|x_{\text{true}}|} \quad (3)$$

where:

- E_r is the relative error
- x_{true} is the true value
- x_{approx} is the approximated value

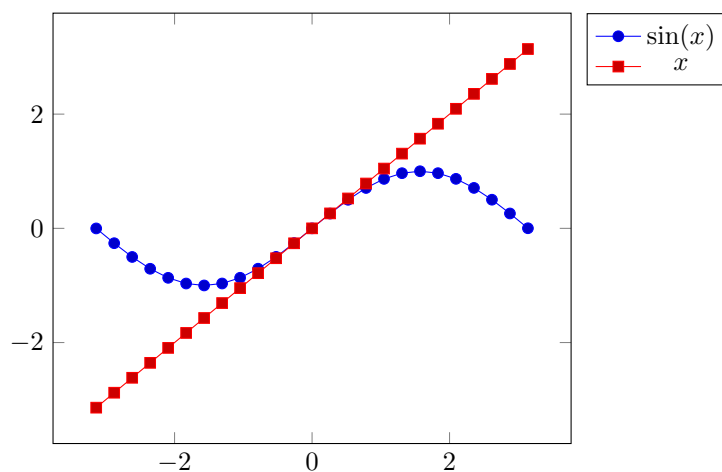
Relative error is often expressed as a percentage:

$$E_{\%} = E_r \times 100\% \quad (4)$$

2.3 A Mathematical Example of Error

Let's now do an example approximation problem with actual values that we can use to calculate error. Suppose we are interested in computing $\sin\left(\frac{\pi}{8}\right)$ but we decide that this problem is too difficult and choose to solve a similar, simpler problem. (NOTE: Obviously, this problem is trivial. but using toy problems with known solutions is an excellent way to learn a technique.) We can make several simplifications:

- Instead of the true input $x = \frac{\pi}{8}$, we use $\hat{x} = \frac{3}{8}$.
- Instead of the true function $f(x) = \sin(x)$, we use the small angle approximation (i.e., first term of the Taylor series for $\sin(x)$) so that $\hat{f}(x) = x$.



Notice how $\sin(x) \approx x$ for small values of x .

Now, the true result (to four digits of precision) is given by:

$$y = f(x) = \sin\left(\frac{\pi}{8}\right) \approx 0.3827.$$

If we use the small angle approximation and a truncated value for π , we obtain the approximate result:

$$\hat{y} = \hat{f}(\hat{x}) = \frac{3}{8} = 0.3750.$$

Thus, the absolute error is:

$$E_a = |\hat{f}(\hat{x}) - f(x)| \approx |0.3750 - 0.3827| = 0.0077$$

And the relative error is:

$$E_r = \left| \frac{\hat{f}(\hat{x}) - f(x)}{f(x)} \right| \approx \left| \frac{0.3750 - 0.3827}{0.3827} \right| = 0.0201$$